

AI and Physics Assisted Character Pose Authoring

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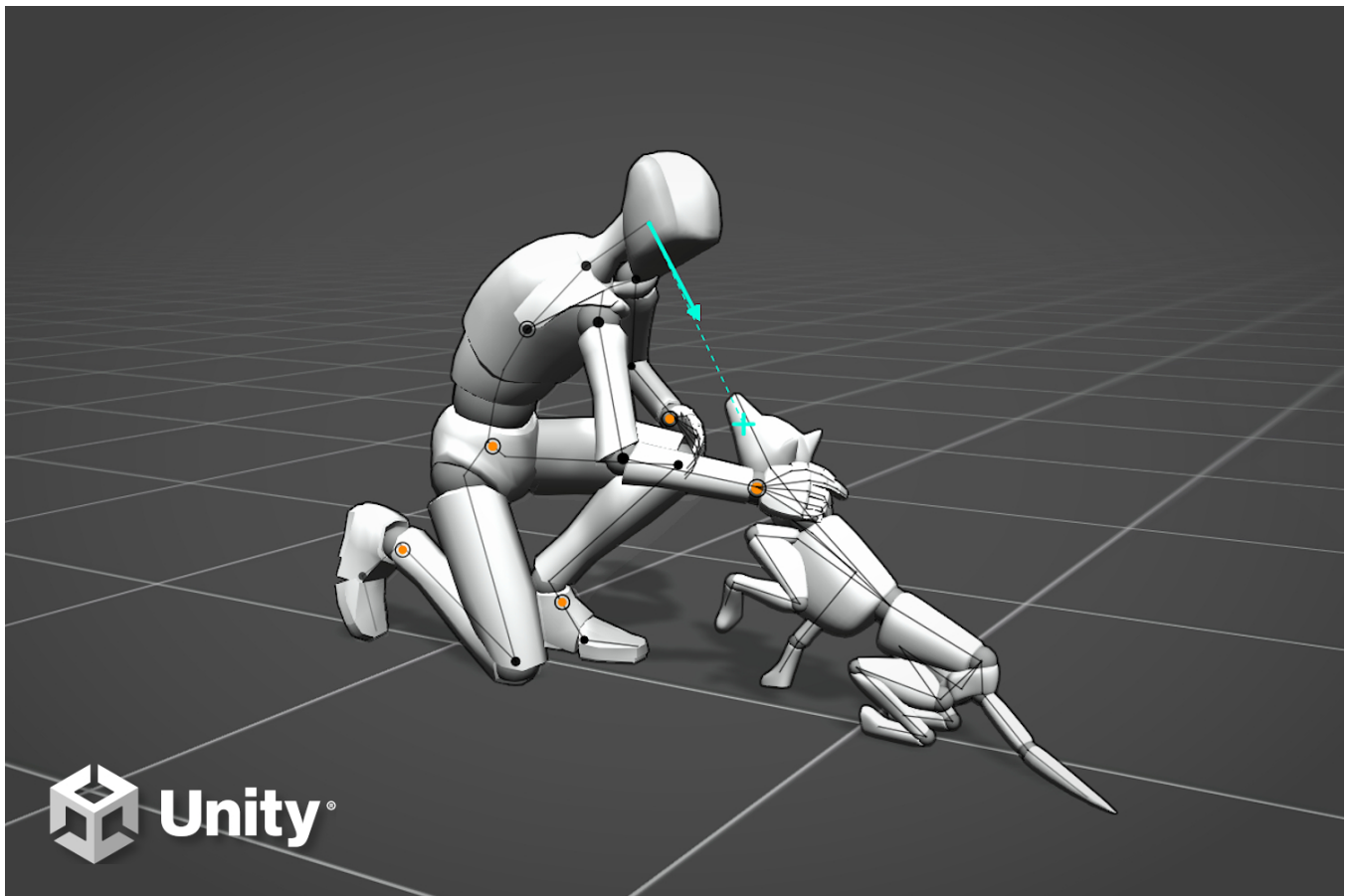


Figure 1: Preview of our character posing tool. (c) 2022 Unity Technologies.

ABSTRACT

We present a tool that allows users to quickly author character poses. Authoring character poses is typically done by professional artists and is a time consuming process that involves a lot of user manipulations. Our tool leverages both machine learning and a physics engine to enable users with no artistic experience to author natural-looking poses in a few seconds.

First, we trained a machine learning (ML) model to predict a full character pose, including individual fingers, from a set of sparse constraints. These constraints allow the user to control the final pose by specifying final joint positions, orientations or a target that they should face. Our ML architecture allows the constraints to be given in any order and number. The model was trained on a large set of motion capture data so that it predicts natural and realistic human poses.

Second, we integrated our ML model with a physics solver so that the final pose also respects environmental constraints such as colliding with other objects. This allows the user to quickly pose a character interacting with the environment, another character or itself.

Finally, we developed a user-friendly interface to control this tool. We believe that the combination of machine learning and physics lower the entry bar to character animation.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning.**

KEYWORDS

animation, posing, character, machine learning, physics, inverse kinematics

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